JNTUH USED 11-05-18AM Set 2

Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
DESIGN PATTERNS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	What is the meaning of "Intent" in design pattern?	[2]
b)	How to use a design pattern? Write the guidelines.	[3]
c)	What is the rationale behind Prototype design pattern?	[2]
d)	List different types of creational patterns.	[3]
e)	Define Bridge pattern.	[2]
f)	Sketch the structure of composite pattern.	[3]
g)	List the benefits of Iterator pattern.	[2]
h)	What is the intent of Memento pattern?	[3]
i)	Write about the purpose of Behavioral patterns.	[2]
j)	Sketch the class diagram for TCP connection class.	[3]

PART-B

(50 Marks)

2.a) Write the general template used for describing the design patterns.
b) List and explain different ways of organizing design patterns. [5+5]

OR

3.a) Give an overview of organizing a catalog.
b) Name the patterns that are included in catalog of design patterns along with their intents. [5+5]

4.a) Explain the Spelling Checking and Hyphenation design problem in detail.
b) Why the Singleton uses a special method to instantiate objects? Discuss. [5+5]

OR

5.a) Discuss about the supporting multiple window systems design problem.
b) Explain the consequences of the Abstract Factory pattern and Builder patterns. [5+5]

6.a) Discuss about the participants of Adapter pattern and explain the functions of each.
b) What are the implementation issues of Façade pattern? Explain with sample code. [5+5]

OR

7.a) Explain about the intent, uses and related patterns of Decorator and Proxy patterns.
b) State the uses and related patterns of Flyweight and Bridge design patterns. [5+5]

8.a) What are the issues to be considered while applying Observer pattern?
b) What are the consequences of Chain of Responsibility pattern? Explain. [5+5]

OR

9.a) Describe the implementation issues of Mediator pattern.
b) Write about the structure and participants of Interpreter pattern. [5+5]

10.a) Discuss about consequences and implementation issues of State design pattern.
b) Explain the participants, collaborations and sample code of Template method. [5+5]

OR

11.a) Write about the intent, motivation, structure and applicability of Visitor pattern.
b) Explain the implementation issues of Strategy pattern and write a sample code. [5+5]

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